

SAPATH · GATE CS 2027

MASTER PLAN — VERSION 10 ELITE

AIR <50 Execution System

17-Layer Elite Operating System

July 2026 – February 2027

17

Elite Layers

2640+

Question Target

38+

Tests + Mocks

AIR <50

Goal

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PART 0 — TEST STRUCTURE & STUDY RULES

Marking Scheme (Official GATE)

Type	Marks	Correct	Incorrect	Unattempted
MCQ 1-mark	1	+1	−0.33 (1/3)	0
MCQ 2-mark	2	+2	−0.67 (2/3)	0
MSQ 1-mark	1	+1 (all correct options only)	0	0
MSQ 2-mark	2	+2 (all correct options only)	0	0
NAT 1-mark	1	+1 (if within accepted range)	0	0
NAT 2-mark	2	+2 (if within accepted range)	0	0

Test Format Overview

Test Type	Frequency	Questions	Duration	Scope
Weekly Test	Every Sunday	25Q	60 min	15Q new + 5Q prev week + 5Q PYQ
Revision Test	Every 2nd Tuesday	50Q	120 min	Cumulative — 50% easy / 30% med / 20% hard
Monthly Test	End of month	65Q / 100 marks	180 min	GATE-standard: 35 MCQ + 15 MSQ + 15 NAT
December Mocks	Every 5 days (7 total)	65Q / 100 marks	180 min	Full syllabus, rotating emphasis
January Mocks	Every day (31 total)	65Q / 100 marks	180 min	Full syllabus, rotating emphasis

The 11 Sacred Rules

Rule 1: Every PYQ solved ≥ 3 times. One in every 5 PYQs repeats in pattern — treat each as a template, not a one-off question.

Rule 2: Daily error log + same-day revision from the log. Zero exceptions. If you don't log it, you will repeat it.

Rule 3: Notes must be concise enough to revise an entire subject in a single sitting. Verbose notes are useless under time pressure.

Rule 4: Spend MORE time analysing mocks than taking them. Minimum ratio — 1 hour mock : 1.5 hours analysis.

Rule 5: Every subject revised at least 3 times across the full plan. First pass = learning. Second pass = solidifying. Third pass = exam-ready.

Rule 6: Maximise speed AND accuracy simultaneously using isolated timed drills every study day.

Rule 7: Every Sunday: mixed PYQ-only session from all completed topics. Full exam conditions. No hints.

Rule 8: Make as many mistakes as possible in practice. High volume, high frequency — mistakes in practice = marks saved in exam.

Rule 9: Build knowledge as a connected web. Every new concept must link to something already known. Isolated facts are forgotten.

Rule 10: Monthly question-solving target increases every month (see Layer 2 numerical targets).

Rule 11: Non-negotiable sacrifice: social media, random internet browsing, excessive sleep. Every hour counts.

PART 1 — AUDIT SUMMARY: v8 CORRECTIONS FROM v6

Engineering Mathematics

- ★ Graph theory: connectivity, matching, colouring — absent from v6 entirely
- ★ Combinatorics: counting, recurrence relations, generating functions — absent from v6 entirely

Programming and Data Structures

- ★ Recursion — official syllabus lists it as its own line item; v6 never scheduled it

Digital Logic

- ★ Combinational circuit building blocks: adders, multiplexers, decoders, encoders (v6 had K-maps only)
- ★ Fixed-point arithmetic — v6 named IEEE-754 floating point but not fixed-point, both required

Computer Organization and Architecture

- ★ Interrupt-driven I/O — v6 covered DMA but not interrupt mode; syllabus names both explicitly

Compiler Design

- ★ Constant propagation — v6 named liveness and CSE but dropped constant propagation

Databases

- ★ Relational algebra and tuple calculus — v6 jumped from relational model straight to SQL
- ★ Integrity constraints — never named as a distinct topic in v6

Computer Networks

- ★ Fragmentation — v6 covered IP addressing/CIDR but not fragmentation (same syllabus clause)

General Aptitude — Pass 1

- ★ Reading comprehension and narrative sequencing — entire Verbal Aptitude sub-topic missing
- ★ Data interpretation: bar/pie charts, 2D/3D plots, maps, tables — missing
- ★ Mensuration and geometry — missing
- ★ Permutations and combinations — missing
- ★ Powers, exponents, logarithms — missing

General Aptitude — Pass 2 (finer sub-clauses)

- ★ Conditional probability — syllabus says 'Conditional probability and Bayes theorem'; v6 kept only Bayes'
- ★ Elementary statistics and probability — distinct GA Quantitative sub-clause, separate from DI
- ★ Spatial Aptitude: cutting and patterns in 2D/3D — v6 kept paper folding but dropped these siblings

Already fully matching official syllabus (no changes made):

Linear Algebra · Calculus · Probability & Statistics · all of Algorithms · all of Theory of Computation (RE/DFA/NFA, CFG/PDA, pumping lemma, Turing machines) · parsing/SDT/runtime-environments/intermediate-code in Compiler Design · all of Operating Systems · normal forms/file-organization/indexing/transactions in DBMS · layering/switching/data-link/routing/transport/application layers in Computer Networks · Analytical and Spatial Aptitude.

PART 2 — 17 ELITE LAYERS: THE AIR <50 OPERATING SYSTEM

LAYER 1 — MASTERY DEFINITION

A topic is NOT done when you finish reading it. It is done only when every criterion in the table is met.

Subject / Area	Mastery Criteria — ALL must be met before moving on
Algorithms — DP	Derive recurrence without hints · Solve medium DP ≤ 8 min · Identify DP pattern ≤ 30 sec · Explain state transitions verbally · Name 3+ pitfalls · Score $\geq 75\%$ on DP-only test
Algorithms — Graph	Trace BFS/DFS manually on paper · Prove MST correctness verbally · Implement Dijkstra from memory · Solve graph PYQs $\geq 80\%$
Theory of Computation	Construct DFA/NFA from RE ≤ 5 min · Build CFG for any described language · Apply pumping lemma to prove non-regularity · Distinguish decidable/undecidable in 1 read
Compiler Design	Draw full parse tree for ambiguous grammar · Compute FIRST/FOLLOW ≤ 4 min · Build SLR/LALR table · Trace 3-address code generation end-to-end
Operating Systems	Solve Banker's algorithm numerically · Trace page-replacement algorithms · Calculate turnaround/waiting time for any scheduling policy · Explain critical section conditions verbally
DBMS	Write relational algebra for any SQL-equivalent query · Normalise to BCNF ≤ 5 min · Compute FD closure · Trace 2PL and detect deadlock
COA	Trace pipeline with all hazard types · Calculate CPI for multi-stage pipeline · Decode instruction formats · Explain cache mapping with numerical example
Computer Networks	Calculate subnet/CIDR · Trace distance-vector convergence · Explain TCP 3-way handshake step-by-step · Calculate sliding window throughput
Digital Logic	Minimise any 4-var function via K-map ≤ 3 min · Design a full-adder circuit from scratch · Trace flip-flop state table · Convert fixed/floating-point correctly
Engineering Math	Solve eigenvalue problem ≤ 6 min · Apply Bayes' theorem numerically · Derive recurrence relation solution · Compute generating functions for combinatorics problems
General Aptitude	Solve any GA question ≤ 90 sec · Read a comprehension passage ≤ 2 min · Interpret 2D/3D DI chart ≤ 2 min

LAYER 2 — QUESTION TARGETS (Numerical Minimums)

These are minimums, not ceilings. Monthly totals escalate to build speed and volume habits.

Subject	Min PYQs	Original Qs	Revision Qs	Hard Qs	Monthly Total
Engineering Math	60	80	60	30	230
Programming / DSA	50	90	70	40	250
Algorithms	70	100	80	40	290
Theory of Computation	60	80	60	30	230
Compiler Design	50	70	50	25	195
Digital Logic	55	75	60	30	220
COA	60	85	65	35	245

Subject	Min PYQs	Original Qs	Revision Qs	Hard Qs	Monthly Total
Operating Systems	65	90	70	40	265
DBMS	65	90	70	35	260
Computer Networks	60	85	65	35	245
General Aptitude	80	60	50	20	210
TOTAL	675	905	700	360	2640

Monthly Escalation Targets:

Month	Min Questions	Cumulative Total	Primary Focus
July 2026	300	300	EM + DSA + Algorithms fundamentals
August 2026	400	700	Algorithms (complete) + TOC
September 2026	500	1,200	Digital Logic + CN (intro)
October 2026	550	1,750	DBMS + CN (complete)
November 2026	600	2,350	COA + OS + Compiler 100%
December 2026	700	3,050	Full mock phase — all subjects
January 2027	900	3,950	Daily mock phase — peak volume

LAYER 3 — ERROR LOG SYSTEM

Every wrong answer must be logged within 1 hour of the test. Use the 7-category taxonomy:

Code	Type	Root Cause	Elimination Protocol
C	Concept mistake	Did not understand the underlying principle	Re-study topic from a different source; add to retrieval practice queue
F	Formula mistake	Applied wrong formula or mis-remembered	Add to Formula Book; review the formula 3× the next day
R	Reading mistake	Misread question stem or options	Re-read every question twice before answering; build the habit
T	Time mistake	Spent too long or ran out of time	Add to Speed Drill list; practice 10 similar questions timed
S	Silly mistake	Arithmetic slip, sign error, unit confusion	Create a personal checklist; 30-sec review before submitting answer
G	Guessing mistake	Guessed on MCQ and attracted negative marks	Apply the 60% confidence rule — if unsure, skip MCQ
M	Memory mistake	Could not recall a memorised fact or definition	Add to spaced-repetition deck immediately; R1 in 2 days

Weekly Error Review Process:

- Every Sunday morning: tally error-type counts for the past week.
- If any single error type appears 3+ times in a week → mandatory 30-min targeted drill that same day.
- Monthly: calculate error-type percentages. The highest category becomes that month's #1 focus area.
- A mistake is 'eliminated' only when you answer 3 consecutive similar questions correctly without reference.

LAYER 4 — SPACED REVISION ALGORITHM

Based on Ebbinghaus forgetting curve. Every topic follows this exact schedule:

Pass	Interval	What to Review	Time Budget	Trigger
R0 — Same day	Day of first study	Full notes + 5 practice questions	20 min	Mandatory — do not skip
R1 — 2 days	2 days after R0	Closed-book recall → check notes → fix gaps	15 min	Scheduled in planner
R2 — 7 days	7 days after R0	Solve 3 PYQs + re-read error log for this topic	20 min	Scheduled in planner
R3 — 21 days	3 weeks after R0	Blank-page recall of all formulas + key concepts	25 min	Scheduled in planner
R4 — 45 days	6 weeks after R0	Mini-test (10Q, timed) + update Formula Book	30 min	Scheduled in planner
R5 — 90 days	3 months after R0	Full subject sweep — notes + 5 PYQs + error log	40 min	Scheduled in planner
R6 — Final	January 1–15	All subjects: rapid-fire 2Q per topic minimum	Full session	Fixed date

Maintain a Revision Calendar. Every topic gets a row; columns are R0–R6. Mark date completed. A missed revision is the first task the next morning — no exceptions.

LAYER 5 — RETRIEVAL PRACTICE PROTOCOL

After completing every topic, execute this full retrieval sequence before moving on to the next topic. Total time: ~40 minutes.

Step	Activity	Time	Purpose
1	Closed-book recall — write everything you remember about the topic on blank paper	15 min	Forces active reconstruction; exposes hidden gaps
2	Blank-page formula dump — write every formula, theorem, and definition from memory	5 min	Tests formula retention without hint of notes
3	Teach-yourself session — explain the topic out loud as if teaching a student	10 min	Fluency gaps become obvious when verbalised
4	Formula reconstruction — derive key formulas from first principles	5 min	Builds deep understanding; not just surface recall
5	Diagram/flowchart reconstruction — draw architectures, state machines, or flowcharts from memory	5 min	Spatial memory reinforces abstract concepts
6	Memory dump review — compare written dump to notes; circle every gap	5 min	Gaps go directly into error log as type M

Research basis: The testing effect (Roediger & Karpicke, 2006) shows retrieval practice produces 50% better long-term retention than re-reading the same material.

LAYER 6 — WEAKNESS DETECTION SYSTEM

Run this diagnostic every Sunday after the PYQ session. Takes 20 minutes. Every dimension must be checked.

Dimension	Warning Signal	Action Threshold	Response
Subject accuracy	% correct per subject (last 3 tests)	Below 60% on any subject	+1 hour/day dedicated to that subject for 2 weeks
Chapter accuracy	% correct per chapter in error log	Below 55% in any chapter	Schedule targeted chapter re-study within 3 days
Concept gaps	Concepts appearing in error log more than twice	Any concept 3+ times in 2 weeks	Add to daily 15-min concept drill for 1 week
Question type	MCQ / MSQ / NAT accuracy split	NAT accuracy below 70%	Daily 10-min NAT drill; MSQ below 65% → same
Difficulty level	Easy / Medium / Hard accuracy breakdown	Easy below 85% OR Hard below 40%	Recalibrate — more easy drills OR hard-only sessions
Speed	Average time per question by type	Over 90 sec average for easy MCQ	Speed drill: 20 easy MCQs in 25 min target
Confidence calibration	Marked 'confident' but wrong ratio	Over 20% of confident answers wrong	Slow down; apply elimination before every answer

LAYER 7 — SPEED DEVELOPMENT DRILLS

Speed and accuracy are trained separately. Combining them in the same drill improves neither.

Speed Type	Target	Drill Method	Frequency
Reading speed	Read 100-word problem in under 25 seconds	Timer drill: 20 questions read-only, no solving	3× per week
Calculation speed	2-digit multiplication under 3 seconds	Mental maths: 50 timed calculations per session	Daily 5 min
Elimination speed	Reject 2 wrong MCQ options in under 15 seconds	Practise elimination-first on every MCQ drill set	Every MCQ session
Formula recall speed	Any standard formula in under 5 seconds	Flashcard speed runs: 30 cards in 3 minutes	Daily 3 min
NAT solving speed	Easy NAT under 3 min; medium NAT under 5 min	Isolated NAT sets: 10 NATs in 30 min target	2× per week
DI scanning speed	Extract key data from chart/table in under 20 sec	5 charts, 3 questions each, timed	2× per week

Per-question time budget (exam day targets):

Question Type	Target Time	Maximum Budget	Action if Over Budget
Easy MCQ — 1 mark	45 seconds	75 seconds	Mark for review, move on
Medium MCQ — 2 mark	90 seconds	2.5 minutes	Mark for review, move on
Hard MCQ — 2 mark	2 minutes	3 minutes	Skip entirely in first pass

Question Type	Target Time	Maximum Budget	Action if Over Budget
MSQ — any	90 seconds	2.5 minutes	Only attempt if $\geq 80\%$ confident on ALL options
Easy NAT — 1 mark	2 minutes	3 minutes	Always attempt — no negative marking
Medium NAT — 2 mark	3 minutes	5 minutes	Always attempt — rough estimate beats blank
Calculation-heavy NAT	4 minutes	5 minutes	Skip if still unsure after 4 min; return last
GA question	60 seconds	90 seconds	Never skip — GA is free marks

LAYER 8 — MOCK TEST ANALYTICS FRAMEWORK

Spend at least 1.5× the mock duration on post-mock analysis. A mock not fully analysed is a mock wasted.

Metric	Definition / Formula	Benchmark to Hit
Raw Score	Sum of correct marks minus all penalty marks	≥55/100 by Dec; ≥70/100 by Jan 3rd week
Accuracy Rate	$\text{Correct} \div \text{Attempted} \times 100$	≥80% overall; MCQ ≥82%; NAT ≥75%
Attempt Rate	$\text{Attempted} \div \text{Total Questions} \times 100$	≥85% by January
Negative Mark Rate	$\text{Negative marks} \div \text{MCQ marks attempted} \times 100$	Below 8%; if over 12% → enforce skip-if-unsure rule
Subject Accuracy Vector	Accuracy % per subject (11 separate values)	No subject below 60% by November
Difficulty Distribution	Easy / Medium / Hard attempt and accuracy split	Attempt ≥90% easy; ≥70% medium
Time Efficiency Score	Questions solved per 10 minutes	≥3.5 Q/10 min by January
Confidence Calibration	% of 'I am sure' answers that were correct	Above 85%; below 70% → slow down globally
Careless Error Count	Silly + Reading mistakes per mock	Below 4 per mock by December
Knowledge Gap Index	$\text{Concept mistakes} \div \text{Total wrong} \times 100$	If over 40% → insufficient revision; trigger R5 cycle
AIR Estimate	Map score to AIR via Layer 16 benchmark table	See Layer 16

Mandatory post-mock protocol (in order):

- Step 1 (30 min): Reattempt every wrong question without time pressure. Categorise each error by type.
- Step 2 (20 min): Calculate all 11 metrics above. Record every value in your dashboard.
- Step 3 (15 min): Identify the single worst subject and the single worst question type.
- Step 4 (10 min): Write a 3-sentence note — what went wrong, why, what changes next mock.
- Step 5 (next morning): 45-min session on the worst subject identified in Step 3.

LAYER 9 — EXAM TIME MANAGEMENT STRATEGY

Three-pass strategy. Never attempt in question-number order.

Pass	Time Window	What to Attempt	Skip Rule
First pass	0 – 90 min	All easy MCQs + all GA questions + easy NATs (1-mark)	Skip anything that needs more than 75 sec
Second pass	90 – 150 min	Medium MCQs, MSQs you are $\geq 80\%$ confident on, medium NATs	Skip anything uncertain after 2.5 min
Third pass	150 – 180 min	Hard MCQs (elimination only), remaining NATs (guess if needed)	Never leave a NAT blank; never guess MCQ below 50%

- MSQ rule: only attempt if you are $\geq 80\%$ confident of ALL correct options. One wrong option = 0 marks, not negative.
- NAT rule: always attempt. No negative marks. Even a rough estimate improves expected score over a blank.
- GA strategy: attempt all GA questions in the first pass. These are the highest-reliability marks.
- Flagging: use the on-screen flag feature for questions you skip. Second-pass begins by reviewing flagged questions only.

LAYER 10 — FORMULA BOOK SYSTEM

Category	What Goes IN	What NEVER Goes In
Formulas	Every formula you have gotten wrong at least once in practice	Formulas you can derive from scratch in under 30 seconds
Definitions	Testable definitions (BCNF conditions, Chomsky Normal Form, MST properties)	General explanations or prose notes
Complexity	Time/space complexity of every standard algorithm	Proof sketches or derivations
Recurrences	Recurrence relations for standard divide-and-conquer algorithms	Things that are better as diagrams
Mnemonics	Every personal mnemonic you have created	Other people's mnemonics you haven't internalised
Inequalities	Master theorem cases, CPI formulas, bandwidth-delay product	Results you can re-derive quickly

Review schedule:

- Add new entries immediately after any error-log type F (Formula mistake).
- Review: 10 minutes every single morning before starting study — rapid-fire recall, no skipping.
- Full review: every Revision Test day, before the test begins.
- Final sprint: January 1–15, review the entire formula book daily. By Jan 15, review time should be under 20 minutes.

LAYER 11 — SUBJECT-SPECIFIC MEMORY STRATEGIES

Compiler Design

- Parsing hierarchy mnemonic: 'Lean Silly Camels Like Afternoon Lounging' → Lexical, Syntax, CFG, LR, LALR, Attribute
- Spaced repetition for FIRST/FOLLOW rules: flashcard drill using 5 different grammars per session
- Memory palace: walk through a 'compiler factory' — front door = lexer, conveyor = parser, assembly = code gen
- Interleave with TOC (both use automata theory) to reinforce cross-connections

Operating Systems

- Deadlock conditions mnemonic: 'My Hamster Never Waits Calmly' → Mutual exclusion, Hold and wait, No preemption, Wait circular
- Scheduling: practice numerical problems daily — do not just memorise algorithms, compute actual values
- Page replacement: create story mnemonics for OPT, LRU, LFU, FIFO as characters making decisions
- Active recall: after reading any OS concept, close notes and explain it out loud to the wall

Computer Networks

- OSI layers: 'All People Seem To Need Data Processing' (Application → Physical)
- TCP congestion control: draw the sawtooth diagram from memory daily for 2 weeks — makes it automatic
- Routing algorithms: connect to graph algorithms you know — Dijkstra = link-state, Bellman-Ford = distance-vector
- Subnetting: 5 CIDR problems every morning during CN month until the process is instantaneous

DBMS

- Normal forms: '1NF no repeating groups · 2NF full FD on PK · 3NF no transitive · BCNF every determinant is a candidate key'
- Relational algebra: link each operator to its SQL equivalent — retrieval via familiar analogy
- 2PL: trace lock/unlock sequences on 3 different transaction sets from memory — repeat 5× during DBMS month
- Interleave with Mathematics: FD closure is set closure — same concept, different notation

COA

- Pipeline stages mnemonic: 'If It Executes Well Memory Works' → IF, ID, EX, MEM, WB
- Cache: draw set-associative mapping diagram from memory — do this 10 times across the COA study month
- Addressing modes: create a one-page table from memory; test yourself every morning for 1 week
- Interrupts vs DMA: DMA = 'bulk shipper' (CPU not involved), Interrupt = 'urgent courier' (CPU handles)

Theory of Computation

- Chomsky hierarchy: draw the concentric circles ($\text{Regular} \subset \text{CFL} \subset \text{CSL} \subset \text{Recursive} \subset \text{RE}$) from memory
- Pumping lemma: practise the 3-step proof structure until automatic — choose string, pump, derive contradiction
- Closure properties: flashcard drill — 'Is CFL closed under intersection? No. Under union? Yes. Complement? No.'
- Turing machine: connect to algorithm design — TM is an abstract computer; decidability = algorithm exists

Engineering Mathematics

- Eigenvalues: derive characteristic equation from scratch every time — do not memorise without understanding
- Probability: Bayes' theorem — practise 5 real-world word problems; the formula becomes intuitive not mechanical
- Generating functions: link to Z-transform if familiar — same algebraic structure, different domain
- Combinatorics: build a decision-tree mental model for every 'how many ways' problem

Digital Logic

- K-map minimisation: practise until any 4-variable map takes under 3 min — pure speed drill, 5 maps per session
- Flip-flops: create state-transition table from memory for JK, D, T, SR — 5-minute daily drill during DL month

- Number systems: fixed-point conversion — 10 problems daily for 1 week; it becomes automatic
- Circuit building blocks: build a full adder from scratch on paper; repeat 5 times total across the month

LAYER 12 — PYQ INTELLIGENCE SYSTEM

Do not solve PYQs randomly. Build a systematic intelligence database for every subject.

Subject	High-Frequency PYQ Topics	Observed Trend	Preparation Priority
Algorithms	DP recurrences, MST correctness proofs, graph algorithm complexity	Increasing — 2-mark NATs rising in frequency	Derivation speed; numerical tracing practice
TOC	DFA minimisation, CFL pumping lemma, TM decidability	Stable — same 4–5 concept types repeat every year	3× repetition on each concept type is sufficient
OS	Banker's algorithm, page replacement, CPU scheduling calculations	Numerical dominance increasing since 2022	Calculation drills; formula accuracy under time pressure
DBMS	Normalisation, SQL joins, transaction serializability	Relational algebra questions returning since 2021	Formal RA + tuple calculus — do not skip these
COA	Pipelining CPI, cache miss rate, addressing modes	Pipeline questions growing harder year on year	Multi-hazard problems; combined cache + pipeline
CN	Subnetting, TCP sliding window, CSMA/CD timing	Subnetting appears every single year without exception	10 subnetting problems per week throughout CN month
Compiler Design	LL/LR parsing tables, FIRST/FOLLOW, 3-address code	LALR ≥ 1 question per year since 2018	Table-building speed; full parse tree tracing
Digital Logic	K-map minimisation, flip-flop sequences, number conversion	Fixed-point questions returning since 2022	Fixed + floating-point combined questions
Eng. Math	Eigenvalues, probability, graph theory, combinatorics	Combinatorics increasing in frequency since 2020	Generating functions + recurrence relation solutions
GA	Series completion, reading comprehension, DI charts, P&C	DI and RC becoming heavier each year	Daily 10-min GA timed practice — never skip this

PYQ solve protocol — three mandatory attempts:

- Attempt 1: Full exam conditions — timed, no hints, full negative-marking awareness. Log every error.
- Attempt 2 (≥ 7 days later): Closed-book recall — solve from memory without re-reading the solution.
- Attempt 3 (≥ 21 days later): Speed run — must solve in 50% of original time. Confirms true mastery.
- Every Sunday: pull 10–15 PYQs randomly from completed topic pool. No preview. Full conditions.

LAYER 13 — ADAPTIVE RECOVERY ALGORITHMS

Missing days happens. These protocols prevent schedule collapse without sacrificing long-term retention.

Days Lost	Recovery Protocol	Preserved Non-Negotiables
1 day	Extend next 2 daily sessions by 30 min each. Skip new topics; prioritise revision and error log catch-up.	Error log + Sunday PYQ session
3 days	Compress 3 days into 2. Cut new content by 30%; reinforce via next Revision Test. Extend by 45 min/day for 1 week.	Weekly test (delay max 1 day); PYQ session
1 week	Activate emergency mode: +2 hours/day for 10 days. Drop hardest 20% of new content. Prioritise Algorithms, OS, DBMS above all.	Monthly test (delay max 3 days)
2 weeks	Accept that 1 subject will be lighter. Identify lowest-weightage topic and cap at 60% depth. Add 1 extra mock to compensate.	Core subjects must not be cut: Algorithms, OS, DBMS, CN

Never try to compress a missed week into 2 days. Spread recovery over 10–14 days. Cramming lost content causes burnout and retention collapse — the opposite of the goal.

LAYER 14 — BURNOUT PREVENTION & COGNITIVE HEALTH

Area	Protocol	Warning Signs
Deload days	1 reduced-intensity day (50% load) every 2 weeks. Light PYQ review + formula book. No new content.	3+ errors on previously correct questions; focus fails at 25 min mark
Study blocks	90-minute Ultradian blocks with hard 15-min screen-free break between blocks	Diminishing recall within a block; re-reading the same sentence
Sleep targets	July–Oct: ≥ 7 hr/night · Nov–Dec: ≥ 6.5 hr/night · January: ≥ 7 hr/night · Exam eve: bed by 9:30 PM	Mood deterioration; slow calculation; error rate spike the day after short sleep
Physical activity	30 min non-academic activity daily — walk, physical training, or music. Non-optional.	Increasing irritability; inability to sit still during study
Night cutoff	No study after 10:30 PM. Sleep is when memory consolidation occurs — studying past this point degrades the next day.	Fatigue + next-day recall failure

LAYER 15 — PERFORMANCE DASHBOARD

Track the following metrics. What is not measured cannot be improved.

Daily log (5 min every evening):

Metric	How to Track	Target
Deep work hours	Start/stop timer per uninterrupted study block	≥ 6 hr weekdays; ≥ 8 hr weekends
Questions attempted	Running count in daily log notebook	See monthly escalation table (Layer 2)
Session accuracy	$\text{Correct} \div \text{Attempted}$ at end of each session	$\geq 72\%$ early months; $\geq 82\%$ by January
Error log entries added	Count of new entries added today	High count = good; means you're practising at edge
Revisions completed	Tick R0–R6 calendar for today's due items	Zero missed revisions — reschedule if needed
Formula book additions	Count of new entries added today	Should decrease month-on-month as gaps close

Weekly summary (15 min every Sunday evening):

Metric	Formula / Method	Warning Signal
Weekly accuracy	$\text{Total correct} \div \text{total attempted across all sessions}$	Below 70% → increase revision time next week
PYQ completion rate	$\text{PYQs solved} \div \text{weekly PYQ target} \times 100$	Below 90% → shift next week's schedule to compensate
Error type distribution	Count errors per category from week's log	Top category gets a dedicated 30-min drill
Revision completion rate	$\text{Planned revisions done} \div \text{total planned}$	Below 100% → Monday is revision-only, no new content
Consistency score	$\text{Days with } \geq 4 \text{ hours deep work} \div 7$	Below 5/7 → identify and eliminate the specific obstacle

Monthly review (30 min at end of every month):

Metric	Description	Benchmark
AIR estimate	Monthly test score mapped to historical GATE AIR conversion (see Layer 16)	See Layer 16 table
Subject readiness index	$\text{Mastery criteria met} \div \text{total criteria per subject}$ (from Layer 1)	$\geq 60\%$ per subject by that month's end
Forgetting rate	% of R2+ revisions where recall was below 70% on a quiz	Below 20% is acceptable
Mock score trajectory	Trend line across all scored tests taken to date	Monotonic increase expected — any dip needs diagnosis
Question velocity	$\text{Total questions solved} \div \text{total deep-work hours}$	Should increase month-on-month

LAYER 16 — AIR BENCHMARK TRAJECTORY

Performance benchmarks based on historical GATE CS score-to-AIR conversion and documented preparation velocity of AIR 1–50 rankers.

Month	Expected Score Range	Expected AIR	Required Achievement
July 2026	Monthly test: 30–40/100	AIR 3000+	EM + DSA + Algo fundamentals solid; error log and revision system both active
August 2026	Monthly test: 38–50/100	AIR 1500–3000	TOC + Compiler 40% + Algorithms complete; session accuracy $\geq 68\%$
September 2026	Monthly test: 45–58/100	AIR 800–1500	Digital Logic + CN (partial) complete; speed improving; ≥ 500 total PYQs solved
October 2026	Monthly test: 52–65/100	AIR 400–800	DBMS + CN complete; post-mock analysis habit locked in; ≥ 1200 total questions
November 2026	Monthly test: 60–72/100	AIR 150–400	COA + OS + Compiler 100% done; all subjects revised $\geq 2\times$; no subject below 58%
December 2026	Full mock avg: 65–75/100	AIR 80–150	7 full mocks complete; rigorous post-mock analysis; AIR trending downward
January 2027 (mid)	Full mock avg: 70–80/100	AIR 30–80	Daily mocks; all subjects at ≥ 3 revisions; all speed targets met
GATE 2027 Target	Score: $\geq 75/100$	AIR <50	All 17 layers active; zero concept gaps; calm, strategic exam execution

Approximate score-to-AIR conversion (GATE CS 2023–2026 data):

Score (/100)	≤ 55	58	62	65	68	72	75	78	82+
Approx. AIR	800+	500–800	300–500	200–300	100–200	50–100	25–50	10–25	Top 10

Key insight: Moving from AIR 500 to AIR 100 requires +8 to +12 marks. Moving from AIR 100 to AIR 50 requires only +3 to +5 more marks but demands near-perfect execution — eliminating careless errors and mastering question selection strategy.

LAYER 17 — FINAL AUDIT vs AIR 1–50 RANKER PROFILES

Analysis of documented preparation strategies from GATE CS AIR 1–50 rankers (2019–2026 public interviews, preparation blogs, and IISc/IIT coaching data).

What AIR 1–10 do that others don't	Typical Gap	v9 Mitigation
Solve each PYQ 4–5 times, not 2–3	Most students solve each PYQ once then move on	Layer 12: mandatory 3-attempt protocol with speed-run on attempt 3
Mock analysis takes longer than the mock itself	Students take mocks, briefly scan answers, then move on	Layer 8: 1.5× analysis rule + mandatory 5-step post-mock protocol
Error log reviewed and actioned daily	Errors noted but not systematically eliminated	Layer 3: daily log + weekly trigger at ≥ 3 occurrences
Formula book opened every single morning	Formula books created but rarely reopened after	Layer 10: mandatory 10-min morning review, every day
Retrieval practice before every new topic	Students read then practise — never reconstruct	Layer 5: 6-step retrieval sequence after every topic completion
Speed drills are isolated from content drills	Speed and accuracy trained together = neither improves	Layer 7: six isolated speed drill types with separate targets
Subject accuracy tracked numerically, weekly	Students feel weak somewhere but never quantify it	Layer 6: Sunday weakness detection with numerical thresholds
3 full revisions before mock phase begins	Most students start mocks before completing 2nd revision	Layer 4: R0–R6 spaced system ensures ≥ 3 revisions automatically
General Aptitude treated as a serious scoring area	GA treated as afterthought; 8–12 marks lost here	v9: GA has dedicated targets, PYQ database, and daily 10-min drills
January pace is steady, not a sprint	Students push too hard in Jan; score drops in final week	Layer 14: deload protocol, sleep targets, January = sustainable pace

Risk factors specific to this plan and mitigations:

- Risk: No peer-pressure environment from IIT/coaching batch → Self-imposed weekly dashboard accountability; treat every test as real exam.
- Risk: Parallel research workload (BraTS paper, FL paper) may compress study time → GATE study block 04:30–12:00 is inviolable; Layer 13 recovery protocols handle spill-over.
- Risk: Overconfidence in strong subjects (Algorithms, EM) leading to neglect → Layer 6 quantitative check forces honest assessment; December mock rotation ensures all subjects stay sharp.
- Risk: Non-IIT background creating impostor syndrome → Every metric is objective; AIR <50 is achievable with systematic execution regardless of undergrad institution.

PART 3 — FULL SCHEDULE (v10 · PARALLEL-SUBJECT RESTRUCTURE)

July 1–14 is unchanged from v9 (sequential subject blocks). From **15 July 2026** onward, the plan switches to a **parallel-subject interleaved structure**: multiple subjects now progress simultaneously every month, each advancing by a fixed percentage, rather than one subject being completed before the next begins.

v10 Restructuring Note — Subject Progression Map

Cumulative % of each subject's syllabus completed by end of each period:

Subject	Jul 15-31	August	September	October	November	Dec 1-15
C + DSA + Algorithms	20%	40%	60%	80%	100%	-
Discrete Math + Eng. Math	20%	40%	60%	80%	100%	-
General Aptitude	20%	40%	60%	80%	100%	-
Theory of Computation	33%	66%	100%	-	-	-
Computer Networks	25%	50%	75%	100%	-	-
COA	20%	60%	100%	-	-	-
Operating Systems	-	-	25%	100%	-	-
DBMS	-	-	-	45%	90%	100%
Digital Logic	-	-	-	-	100%	-
Compiler Design	-	-	-	-	-	100%

Note on Linear Algebra: Linear Algebra was already covered in the 12 July test and is excluded from the 'Discrete Math + Eng. Math' percentage above — that bucket now tracks the remaining Discrete Math (graph theory, combinatorics), Calculus, and Probability and Statistics sub-areas only.

Test calendar reshaping: Tests resume on **25 July** (Saturday) as a single hybrid test covering all six subjects' first-stage content — replacing four separate tests (19 Jul, 26 Jul, 28 Jul, 30 Jul) under the old structure. That shortfall of 3 tests is repaid with **3 extra revision tests added in October** (6, 16, 23 Oct). August, September, November, and December dates are unchanged from v9 — only the content mapped into each slot has changed.

Density warning: Compiler Design's entire syllabus is compressed into the first 15 days of December, alongside DBMS's final 10%. This is the tightest window in the whole plan. Begin light pre-reading of Compiler Design in the last week of November if COA/DL/DBMS are on schedule.

JULY 2026 — Engineering Math · Programming · DSA · TOC (intro) · Compiler 20% · Aptitude

Sunday 05 July 2026 — WEEKLY TEST 25Q / 60 min

15Q — This week's new content:

- Discrete Math: propositional & FOL logic
- Sets, relations, functions, partial orders, lattices
- Monoids & Groups
- C: data types, operators, control flow, basic pointers

5Q — Previous week: n/a (first week of prep)

5Q — PYQs from cumulative syllabus taught to date

Sunday 12 July 2026 — WEEKLY TEST 25Q / 60 min

15Q — This week's new content:

- Linear Algebra: matrices, determinants, systems of equations, eigenvalues/eigenvectors, LU decomposition
- Calculus: limits, continuity, differentiability, maxima/minima, MVT, integration
- C: pointers, arrays (1D/2D), address computation

5Q — Medium, previous week's syllabus:

- Discrete Math: propositional & FOL logic · Sets, relations, functions, partial orders, lattices · Monoids & Groups
- C: data types, operators, control flow, basic pointers

5Q — PYQs from cumulative syllabus taught to date

Tuesday 14 July 2026 — REVISION TEST 1 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank

This month so far:

- Discrete Math: propositional & FOL logic · Sets, relations, functions · Monoids & Groups
- C: data types, operators, control flow, basic pointers
- Linear Algebra: matrices, determinants, eigenvalues, LU decomposition
- Calculus: limits, continuity, MVT, integration · C: pointers, arrays

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

JULY 15-31, 2026 - PARALLEL START: 6 subjects begin simultaneously

15-24 July: pure study block, no test. Cover all six subjects' stage-1 content below.

Saturday 25 July 2026 (HYBRID TEST - replaces old Jul19/26/28/30 slots) — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

C + DSA + Algorithms (20%):

- Recursion: base/recursive case, recursion trees, recursive vs iterative
- Stacks and Queues: array + linked-list implementations
- Linked lists: singly/doubly/circular

Discrete Math + Eng. Math (20%):

- Discrete Math: Graph theory (connectivity, matching, colouring)

General Aptitude (20%):

- Verbal: basic English grammar (tenses, articles, prepositions, conjunctions)
- Verbal: basic vocabulary (words, idioms, phrases in context)

Theory of Computation (33%):

- Regular expressions and finite automata (DFA/NFA construction)

Computer Networks (25%):

- Concept of layering: OSI and TCP/IP Protocol Stacks
- Basics of packet, circuit and virtual circuit-switching

COA (20%):

- Machine instructions and addressing modes

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Cumulative % achieved by 31 July: C+DSA+Algo 20% · Discrete+EngMath 20% · GA 20% · TOC 33% · CN 25% · COA 20%

AUGUST 2026 - Parallel Stage 2: six subjects continue

Sunday 02 August 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

C + DSA + Algorithms -> 40%:

- Trees, Binary Search Trees

Discrete Math + Eng. Math -> 40%:

- Discrete Math: Combinatorics (counting, recurrence relations, generating functions)

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Sunday 09 August 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

C + DSA + Algorithms -> 40%:

- Binary heaps, Graph representation (adjacency matrix/list)

COA -> 60%:

- ALU, data-path and control unit

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 11 August 2026 — REVISION TEST 1 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. Covers all stage-1 content plus partial stage-2 progress across all six subjects.

C+DSA+Algo (20%->partial 40%):

- Recursion, stacks, queues, linked lists, trees, BST

Discrete+EngMath (20%->40%):

- Graph theory, Combinatorics

GA (20%):

- Verbal: grammar, vocabulary

TOC (33%):

- Regular expressions and finite automata

CN (25%):

- Layering, switching

COA (20%->partial 60%):

- Machine instructions, addressing modes, ALU/datapath/control unit

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Sunday 16 August 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

Theory of Computation -> 66%:

- Context-free grammars and push-down automata

Computer Networks -> 50%:

- Data link layer: framing, error detection, Medium Access Control, Ethernet bridging

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Sunday 23 August 2026 — WEEKLY TEST 25Q / 60 min**New content this week (multi-subject):****COA -> 100%:**

- Instruction pipelining, pipeline hazards

Computer Networks -> 50% (consolidation):

- Routing protocols: shortest path, flooding, distance vector, link state routing

General Aptitude -> 40%:

- Verbal: reading comprehension, narrative sequencing

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date**Tuesday 25 August 2026 — REVISION TEST 2 50Q / 120 min**

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. COA now complete (100%). TOC at 66%. CN at 50%->progressing to routing.

C+DSA+Algo (40%):

- Trees, BST, binary heaps, graph representation

Discrete+EngMath (40%):

- Graph theory, Combinatorics (complete)

GA (40%):

- Verbal complete: grammar, vocabulary, RC, narrative sequencing

TOC (66%):

- RE/DFA/NFA, CFG, PDA

CN (50%->75%):

- Layering, switching, data link layer, routing protocols

COA (100% COMPLETE):

- Machine instructions through pipelining and hazards

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Thursday 27 August 2026 — MONTHLY TEST 65Q / 100 marks / GATE-standard

35 MCQ + 15 MSQ + 15 NAT. Scope: All six subjects at their August cumulative %: C+DSA+Algo 40% / Discrete+EngMath 40% / GA 40% / TOC 66% / CN 50-75% / COA 100% (complete)

Layer 8 mandatory post-mock analysis (≥90 min). Calculate AIR estimate. Update Layer 15 dashboard.

Sunday 30 August 2026 - Pure PYQ Mixed-Bag Session (no formal marks)

Pull 20+ PYQs from all subjects' content covered to date. Prioritise COA (now complete) for deep practice.

SEPTEMBER 2026 - TOC and COA finish; OS begins

Sunday 06 September 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

C + DSA + Algorithms -> 60%:

- Searching, sorting (elementary + advanced)

COA -> 100% (consolidation):

- Memory hierarchy: cache, main memory and secondary storage

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Sunday 13 September 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

Theory of Computation -> 100%:

- Regular and context-free languages, pumping lemma

Computer Networks -> 75%:

- Fragmentation and IP addressing, IPv4, CIDR notation

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 15 September 2026 — REVISION TEST 1 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. TOC nearing completion. COA fully complete and in revision-only mode.

C+DSA+Algo (60%):

- Searching, sorting added to prior DSA content

TOC (66%->100%):

- CFG, PDA, regular/CF languages, pumping lemma

CN (50%->75%):

- Data link, routing, fragmentation/IP addressing/CIDR

COA (100% (revision only)):

- Full subject — all 5 official bullets

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Sunday 20 September 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

Theory of Computation -> 100% (complete):

- Turing machines and undecidability

COA -> 100% (consolidation):

- I/O interface: interrupt and DMA mode

Computer Networks -> 75% (consolidation):

- Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT)

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Sunday 27 September 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):**C + DSA + Algorithms -> 60%:**

- Hashing (chaining, open addressing)
- Asymptotic worst case time and space complexity

Discrete Math + Eng. Math -> 60%:

- Calculus: limits, continuity, differentiability, maxima/minima, mean value theorem

General Aptitude -> 60%:

- Quantitative: data interpretation (bar/pie charts, 2D/3D plots, maps, tables)

Operating Systems -> 25% (NEW SUBJECT BEGINS):

- System calls, processes, threads, inter-process communication

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 29 September 2026 — REVISION TEST 2 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. TOC and COA both 100% complete. OS newly started at 25%.

C+DSA+Algo (60%):

- DSA complete + searching, sorting, hashing, asymptotic complexity

Discrete+EngMath (60%):

- Graph theory, combinatorics, calculus (limits through MVT)

GA (60%):

- Verbal complete + data interpretation

TOC (100% COMPLETE):

- Full subject — all 4 official clauses

CN (75%):

- Layering through IP support protocols and NAT

COA (100% COMPLETE):

- Full subject — revision only from here

OS (25% (new)):

- System calls, processes, threads, IPC

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Wednesday 30 September 2026 — MONTHLY TEST 65Q / 100 marks / GATE-standard

35 MCQ + 15 MSQ + 15 NAT. Scope: C+DSA+Algo 60% / Discrete+EngMath 60% / GA 60% / TOC 100% (complete) / CN 75% / COA 100% (complete) / OS 25%

Layer 8 mandatory post-mock analysis (≥90 min). Calculate AIR estimate. Update Layer 15 dashboard.

OCTOBER 2026 - CN and OS finish; DBMS begins; +3 EXTRA REVISION TESTS

Densest month in the plan. CN and OS both complete this month while DBMS begins. Three extra revision tests (6, 16, 23 Oct) compensate for July's compression — see v10 Restructuring Note at the start of Part 3.

Sunday 04 October 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

DBMS -> 45% (NEW SUBJECT BEGINS):

- ER-model
- Relational model: relational algebra, tuple calculus, SQL

Operating Systems -> 100%:

- Concurrency and synchronization

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 06 October 2026 — EXTRA REVISION TEST A (October Compensation) 50Q / 120 min

Purpose: Compensates for the 3 tests lost when July compressed into a single 25 July test. Targets the two hardest stage-1 concepts from the single 25 July test, which had no dedicated follow-up.

- C+DSA+Algo: Recursion (base/recursive case, recursion trees) — full re-derivation from scratch
- C+DSA+Algo: Stacks, Queues, Linked lists — closed-book implementation practice
- Discrete Math: Graph theory (connectivity, matching, colouring) — full PYQ set

Layer 8: run full post-test analysis. This is a deep-consolidation pass on compressed July content.

Sunday 11 October 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

DBMS -> 45% (consolidation):

- Integrity constraints

Operating Systems -> 100%:

- Deadlock (detection, prevention, avoidance, Banker's algorithm)

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 13 October 2026 — REVISION TEST 1 (standard) 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. DBMS newly at 45%. OS progressing toward completion.

CN (75%->100%):

- Layering through NAT, now adding transport layer

OS (25%->75%):

- System calls/IPC through concurrency and deadlock

DBMS (45%):

- ER-model, relational algebra/tuple calculus/SQL, integrity constraints

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Friday 16 October 2026 — EXTRA REVISION TEST B (October Compensation) 50Q / 120 min

Purpose: Compensates for the 3 tests lost when July compressed into a single 25 July test. Consolidates the two subjects that finished in September, before December's mock phase begins.

- Theory of Computation — full syllabus PYQ set (DFA/NFA through Turing machines)
- COA — full syllabus PYQ set (machine instructions through I/O interface)

Layer 8: run full post-test analysis. This is a deep-consolidation pass on compressed July content.

Sunday 18 October 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

C + DSA + Algorithms -> 80%:

- Algorithm design: greedy (activity selection, Huffman, knapsack)
- Dynamic programming (LCS, LIS, 0/1 knapsack, matrix chain)

Computer Networks -> 100%:

- Transport layer: flow control and congestion control, UDP, TCP, sockets

Discrete Math + Eng. Math -> 80%:

- Calculus: integration

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Friday 23 October 2026 — EXTRA REVISION TEST C (October Compensation) 50Q / 120 min

Purpose: Compensates for the 3 tests lost when July compressed into a single 25 July test. Consolidates Computer Networks (finishing this month) and General Aptitude's early stages, both of which received lighter weekly-test time during the July compression.

- Computer Networks — full syllabus PYQ set (layering through transport layer)
- General Aptitude — Verbal (grammar, vocabulary, RC, narrative sequencing) full PYQ set

Layer 8: run full post-test analysis. This is a deep-consolidation pass on compressed July content.

Sunday 25 October 2026 — WEEKLY TEST 25Q / 60 min

New content this week (multi-subject):

C + DSA + Algorithms -> 80%:

- Divide-and-conquer (merge sort, Strassen's algorithm, master theorem)

Computer Networks -> 100%:

- Application layer protocols: DNS, SMTP, HTTP, FTP, Email

Operating Systems -> 100%:

- CPU and I/O scheduling
- Memory management and virtual memory
- File systems

DBMS -> 45% (consolidation):

- Integrity constraints (review)

Discrete Math + Eng. Math -> 80%:

- Probability & Statistics: random variables, distributions (uniform, normal, exponential, Poisson, binomial)

General Aptitude -> 80%:

- Quantitative: numerical computation (ratios, %, powers, exponents, logs, P&C, series), mensuration and geometry
- Analytical: logic (deduction, induction), analogy, numerical relations and reasoning

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 27 October 2026 — REVISION TEST 2 (standard) 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. CN and OS both complete. DBMS steady at 45%.

C+DSA+Algo (80%):

- Through divide-and-conquer

Discrete+EngMath (80%):

- Through integration and probability distributions

GA (80%):

- Verbal, DI, numerical computation, mensuration, analytical

CN (100% COMPLETE):

- Full subject — revision only from here

OS (100% COMPLETE):

- Full subject — revision only from here

DBMS (45%):

- ER-model through integrity constraints

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Thursday 29 October 2026 — MONTHLY TEST 65Q / 100 marks / GATE-standard

35 MCQ + 15 MSQ + 15 NAT. Scope: C+DSA+Algo 80% / Discrete+EngMath 80% / GA 80% / CN 100% (complete) / OS 100% (complete) / DBMS 45%

Layer 8 mandatory post-mock analysis (≥90 min). Calculate AIR estimate. Update Layer 15 dashboard.

NOVEMBER 2026 - C+DSA+Algo/Discrete+EngMath/GA finish; Digital Logic complete in one month

Sunday 01 November 2026 — WEEKLY TEST 25Q / 60 min**New content this week (multi-subject):****Digital Logic -> 100% (NEW SUBJECT, finishes this month):**

- Boolean algebra

DBMS -> 90%:

- Normal forms (1NF, 2NF, 3NF, BCNF)

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date**Sunday 08 November 2026 — WEEKLY TEST 25Q / 60 min****New content this week (multi-subject):****Digital Logic -> 100%:**

- Combinational and sequential circuits

DBMS -> 90%:

- File organization, indexing (B and B+ trees)

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date**Tuesday 10 November 2026 — REVISION TEST 1 50Q / 120 min**

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. Digital Logic halfway through its single-month completion. DBMS approaching 90%.

Digital Logic (50%):

- Boolean algebra, combinational and sequential circuits

DBMS (90%):

- ER-model through normal forms and indexing

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Sunday 15 November 2026 — WEEKLY TEST 25Q / 60 min**New content this week (multi-subject):****Digital Logic -> 100%:**

- Minimization (K-maps, SOP/POS)

C + DSA + Algorithms -> 100%:

- Graph traversals (BFS/DFS)
- Minimum spanning trees (Prim's, Kruskal's)

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date**Sunday 22 November 2026 — WEEKLY TEST 25Q / 60 min****New content this week (multi-subject):****Digital Logic -> 100% (COMPLETE):**

- Number representations and computer arithmetic (fixed and floating point)

C + DSA + Algorithms -> 100% (COMPLETE):

- Shortest paths (Dijkstra's, Bellman-Ford)

Discrete Math + Eng. Math -> 100% (COMPLETE):

- Probability & Statistics: mean, median, mode, standard deviation, conditional probability and Bayes theorem

General Aptitude -> 100% (COMPLETE):

- Spatial: transformation of shapes (translation, rotation, scaling, mirroring, assembling, grouping)
- Spatial: paper folding, cutting, and patterns in 2 and 3 dimensions

Remaining questions — PYQs + medium recap from cumulative syllabus taught to date

Tuesday 24 November 2026 — REVISION TEST 2 50Q / 120 min

Cumulative scope — 50% easy / 30% medium / 20% hard — mixed PYQ + AI-generated bank. C+DSA+Algo, Discrete+EngMath, GA, and Digital Logic ALL reach 100% this test.

C+DSA+Algo (100% COMPLETE):

- Full subject — all DSA and Algorithms content

Discrete+EngMath (100% COMPLETE):

- Full subject — Discrete Math, Calculus, Probability & Statistics

GA (100% COMPLETE):

- Full subject — verbal, quantitative, analytical, spatial

Digital Logic (100% COMPLETE):

- Full subject — Boolean algebra through computer arithmetic

DBMS (90%):

- Everything except transactions and concurrency control

Layer 8: run full post-test analysis (all 11 metrics). Log results in dashboard.

Thursday 26 November 2026 — MONTHLY TEST 65Q / 100 marks / GATE-standard

35 MCQ + 15 MSQ + 15 NAT. Scope: C+DSA+Algo 100% / Discrete+EngMath 100% / GA 100% / TOC 100% / CN 100% / COA 100% / OS 100% / Digital Logic 100% / DBMS 90% — only DBMS's final 10% and Compiler Design remain in the whole syllabus

Layer 8 mandatory post-mock analysis (≥90 min). Calculate AIR estimate. Update Layer 15 dashboard.

Sunday 29 November 2026 - Pure PYQ Mixed-Bag Session (no formal marks)

Full-syllabus PYQ sweep across all 8 completed subjects. This is the last PYQ sweep before the December mock phase — prioritise subjects with lowest Layer 15 accuracy.

27-30 November - Compiler Design pre-study window (self-study, not a formal test):

- Lexical analysis, parsing, syntax-directed translation
- Runtime environments

Starting Compiler Design here — before December begins — is what makes the December compression survivable. See the density warning at the start of Part 3.

DECEMBER 1-15, 2026 - DBMS finishes; Compiler Design entirely compressed here

This window uses the original December mock-phase dates (full-syllabus mocks every 5 days). Because December is mock-only by design, all NEW content here is learned in self-study windows between mocks — the table below marks both.

Date	Type	What Happens
01 Dec 2026 (Tue)	MOCK 1 (full syllabus)	Tests everything through Nov + DBMS at 90% + Compiler pre-study (lexical/parsing/SDT, runtime environments)
01-05 Dec	STUDY WINDOW	Compiler Design: Intermediate code generation; Local optimization, data-flow analyses (constant propagation, liveness analysis, common sub-expression elimination)
06 Dec 2026 (Sun)	MOCK 2 (full syllabus)	Compiler Design now substantially complete — tested for the first time
06-11 Dec	STUDY WINDOW	DBMS final 10%: Transactions and concurrency control. Compiler Design full consolidation.
11 Dec 2026 (Fri)	MOCK 3 (full syllabus)	FULL SYLLABUS 100% COMPLETE — every subject, every topic, for the first time

By 11 December 2026, the entire GATE CS 2026 syllabus (all 10 sections + General Aptitude) is complete. Mocks 4-7 (16, 21, 26, 31 Dec) continue unchanged from the original plan — full-syllabus mocks with rotating emphasis, now testing a genuinely complete syllabus.

DECEMBER 16-31, 2026 — CONTINUED MOCK PHASE (Mocks 4-7, unchanged from v9)

Full syllabus is 100% complete by 11 Dec (see above). These four mocks continue the original v9 dates and rotation, now testing a genuinely complete syllabus.

Date	Day	Mock #	Emphasis (every mock remains full-syllabus)
16 Dec 2026	Wednesday	Mock 4	DBMS
21 Dec 2026	Monday	Mock 5	COA + OS
26 Dec 2026	Saturday	Mock 6	Aptitude + mixed cross-subject
31 Dec 2026	Thursday	Mock 7	Balanced — no bias, exact GATE proportion

Post-mock analysis rule (December): each mock must be followed by ≥ 2.5 hours of Layer 8 analysis. Weakest subject from each mock gets a 1-hour targeted session the next morning.

JANUARY 2027 — DAILY MOCK PHASE (1 full mock per day · 31 mocks total)

Full syllabus every day. Emphasis rotates through the same 7-zone cycle (~4.4 cycles). Final 3 days (Jan 29–31) run pure balanced GATE-proportion simulations to mirror exam day.

Date	Day	Mock #	Emphasis
Jan 01	Fri	Mock 1	Engineering Math + Programming/DSA/Algorithms
Jan 02	Sat	Mock 2	TOC + Compiler Design
Jan 03	Sun	Mock 3	Digital Logic + Computer Networks
Jan 04	Mon	Mock 4	DBMS
Jan 05	Tue	Mock 5	COA + OS
Jan 06	Wed	Mock 6	Aptitude + mixed cross-subject
Jan 07	Thu	Mock 7	Balanced — exact GATE proportion
Jan 08	Fri	Mock 8	Engineering Math + Programming/DSA/Algorithms
Jan 09	Sat	Mock 9	TOC + Compiler Design
Jan 10	Sun	Mock 10	Digital Logic + Computer Networks
Jan 11	Mon	Mock 11	DBMS
Jan 12	Tue	Mock 12	COA + OS
Jan 13	Wed	Mock 13	Aptitude + mixed cross-subject
Jan 14	Thu	Mock 14	Balanced — exact GATE proportion
Jan 15	Fri	Mock 15	Engineering Math + Programming/DSA/Algorithms
Jan 16	Sat	Mock 16	TOC + Compiler Design
Jan 17	Sun	Mock 17	Digital Logic + Computer Networks
Jan 18	Mon	Mock 18	DBMS
Jan 19	Tue	Mock 19	COA + OS
Jan 20	Wed	Mock 20	Aptitude + mixed cross-subject
Jan 21	Thu	Mock 21	Balanced — exact GATE proportion
Jan 22	Fri	Mock 22	Engineering Math + Programming/DSA/Algorithms
Jan 23	Sat	Mock 23	TOC + Compiler Design
Jan 24	Sun	Mock 24	Digital Logic + Computer Networks
Jan 25	Mon	Mock 25	DBMS
Jan 26	Tue	Mock 26	COA + OS
Jan 27	Wed	Mock 27	Aptitude + mixed cross-subject
Jan 28	Thu	Mock 28	Balanced — exact GATE proportion
Jan 29	Fri	Mock 29	Balanced — exact GATE proportion (FINAL SIMULATION)
Jan 30	Sat	Mock 30	Balanced — exact GATE proportion (FINAL SIMULATION)
Jan 31	Sun	Mock 31	Balanced — exact GATE proportion (FINAL SIMULATION)

January daily routine:

- 04:30–05:00: Formula book rapid review
- 05:00–08:00: Subject rapid revision (1 subject per morning)
- 09:00–12:00: Full mock (180 min) — exam conditions, no interruptions
- 13:00–15:30: Post-mock analysis — all 11 Layer 8 metrics calculated and logged
- 16:00–18:00: Targeted drill on the weakest subject/topic identified in analysis
- 19:00–20:00: Error log review + spaced-repetition card run
- Bed by 22:30 — non-negotiable in exam month

EXAM WINDOW — FINAL 7 DAYS + GATE DAY

Day	Date	Protocol
Day 1–3 post-mock	Feb 01–03	Light revision only: Formula Book, error log summary, PYQ patterns list. Maximum 4 hours/day. No new content.
Day 4–5	Feb 04–05	Rest-dominant. 2 hours of easy PYQ practice only. Walk, sleep well, eat well. No pressure.
Day 6 (eve of exam)	Feb 06	Zero study after noon. 30-min formula book in morning. Prepare everything the night before. Bed by 9:30 PM.
GATE CS 2027 EXAM	Sunday 07 February	Formula book review 05:00 (30 min). Light breakfast. Arrive 45 min early. Three-pass strategy. Calm execution.

TARGET: AIR <50 · GATE CS 2027

17 Layers Active · All Official Syllabus Topics Covered · Parallel-Subject Restructure Active

v10 Elite — built on v9 Elite — supersedes all previous versions

ROUND 1 CORRECTIONS — TOPIC-EXISTENCE GAPS

Audit 1 found no missing major section. The following specific official clauses were not explicit enough in v10 and are now added.

Digital Logic — minimization methods

Official requirement: Algebraic technique, Karnaugh map, tabular method.

Integrated correction: Add explicit minimization methods: algebraic technique + K-map + tabular method; retain SOP/POS practice.

Schedule placement: November Digital Logic minimization block.

COA — control-unit implementations

Official requirement: Design of control unit — hardwired and microprogrammed.

Integrated correction: Replace generic control-unit wording with explicit hardwired and microprogrammed control coverage, including comparison and design/tracing practice.

Schedule placement: August 09 COA 60% stage-2 block.

COA — memory interfacing and hierarchy performance

Official requirement: Memory interfacing and hierarchy: performance, cache memory mapping.

Integrated correction: Add explicit memory interfacing plus memory-hierarchy performance metrics; retain cache/main/secondary memory study.

Schedule placement: September 06 COA consolidation block.

Computer Networks — performance metrics

Official requirement: Basics of switching (circuit, packet and virtual circuit) and performance metrics.

Integrated correction: Add explicit switching/network performance metrics: latency, bandwidth, throughput, delay and utilization calculations.

Schedule placement: October 18/25 CN completion blocks.

ROUND 1 CORRECTIONS — GENERAL APTITUDE GRAMMAR

The official GA source enumerates grammar components that v10 did not name individually. These are now explicit.

GA — grammar: adjectives

Official requirement: Basic English grammar includes adjectives.

Integrated correction: Explicitly add adjectives to the July grammar syllabus.

Schedule placement: July 25 hybrid-test new-content block / GA 20% stage-1.

GA — grammar: verb-noun agreement

Official requirement: Basic English grammar includes verb-noun agreement.

Integrated correction: Explicitly add verb-noun agreement with targeted identification/correction drills.

Schedule placement: July 25 hybrid-test new-content block / GA 20% stage-1.

GA — grammar: other parts of speech

Official requirement: Basic English grammar includes other parts of speech.

Integrated correction: Explicitly add nouns, pronouns, adverbs, verbs, interjections and related parts-of-speech recognition/correction under the existing grammar block.

Schedule placement: July 25 hybrid-test new-content block / GA 20% stage-1.

AUDIT 1 IMPLEMENTATION RULE

These additions are **binding**. In all subsequent audits, a topic counts as present only when its explicit clause-level coverage can be located either in the original v10 plan or in this corrective addendum. The addendum does not replace any existing schedule content; it supplements it.

Round 1 result after correction: all official parent sections and the previously flagged specific topic-existence items are now explicitly represented in the strategy.

ROUND 2 CORRECTIONS - WORD-BY-WORD CLAUSE GAPS

Audit 2 was run against the official GATE 2027 CS syllabus and the GATE 2026 General Aptitude syllabus. Four clauses were not explicit enough in v10.1. They are now inserted as binding syllabus and schedule entries.

Official clause	Status before A2	Binding correction + schedule placement
Engineering Mathematics - system of linear equations	Partial / generic wording	Explicitly name systems of linear equations in the Linear Algebra syllabus and July 12 test scope. Include solving systems and matrix-form representation in revision/retrieval.
Computer Networks - socket API	Partial / "sockets" only	Replace generic transport "sockets" wording with explicit socket API coverage. Add API purpose, client/server flow, addressing and basic use in the Oct 18 transport-layer block and all later cumulative tests.
GA Quantitative - other graphs representing data	Partial	Explicitly add other graphs representing data to the DI syllabus alongside bar/pie charts, plots, maps and tables; include it in the Sep 27 DI block and cumulative GA tests.
GA Quantitative - estimation	Partial / computation only	Explicitly add numerical computation and estimation , with estimation/approximation practice, to the Sep/Oct quantitative progression and all later full-syllabus testing.

Implementation rule: these four clauses are now explicit requirements in the strategy. They are not merely audit notes.

ROUND 2 - TEST / MOCK SCHEDULE INTEGRITY CORRECTIONS

Test-scope finding 1: The December sequence labels Mock 1 and Mock 2 as "full syllabus" even though their stated scopes are still incomplete. Mock 1 is through November + DBMS 90% + Compiler pre-study; Mock 2 occurs before the DBMS final 10% study window is complete.

Binding correction: Reclassify Dec 1 and Dec 6 as **full-syllabus-format pre-completion mocks**, not literal 100% syllabus mocks. Dec 11 Mock 3 is the first literal 100% syllabus mock because DBMS final 10% and the full Compiler content have been completed by then.

Test-scope finding 2: Scope labels such as "60%", "75%" or "80%" can conceal individual clauses. The corrected plan now requires each percentage scope to be interpreted against the official clause ledger, with the newly added A2 clauses carried into the relevant cumulative tests.

ROUND 2 - CHRONOLOGICAL TEST COVERAGE AUDIT

All scheduled assessments from July through the final January simulations were checked for progression integrity. Early tests are allowed to be partial by design; the audit requirement is that each new block is assessed and that cumulative tests do not silently drop previously taught official clauses.

Period	Assessment block	Audit result	Required correction
Jul	Jul 5 / Jul 12 weekly + Jul 14 revision	Pass	Jul 12 explicitly carries "system of linear equations" after A2 correction.
Jul	Jul 25 hybrid test	Pass with A1/A2 carry-through	GA grammar remains explicit; cumulative clause ledger governs later tests.
Aug	Aug weekly + Aug 11/25 revision + Aug 27 monthly	Pass	COA control variants and later cumulative clauses remain bound to the ledger.
Sep	Sep 6/13/20/27 + Sep 15/29 revision + Sep 30 monthly	Pass with A2 hardening	Sep 27 DI explicitly includes other graphs representing data; GA quantitative progression carries estimation.
Oct	Oct 4/11/18/25 + Oct 6/13/16/23/27 + Oct 29 monthly	Pass with A2 hardening	Oct 18 explicitly includes socket API; Oct 25 quantitative block explicitly includes estimation.
Nov	Nov 1/8/15/22 + Nov 10/24 revision + Nov 26 monthly	Pass	Full-subject tests carry every completed clause; DBMS final 10% remains correctly identified as unfinished.
Dec	Dec 1/6/11 + Dec 16/21/26/31 mocks	Pass after relabel	Dec 1/6 = pre-completion full-format mocks; Dec 11 = first literal 100% syllabus mock; Mocks 4-7 are genuinely full-syllabus.
Jan	Jan 1-31 daily mocks + final simulations	Pass	Every January mock is full-syllabus by construction after Dec 11 completion.

Round 2 final determination: after these corrections, no official clause remains missing from the planned study/test pipeline. The only test-schedule contradiction was the literal "full syllabus" label on Dec 1 and Dec 6; that wording is corrected above.

The original strategy contains 7 December full-syllabus-format mocks, but only the Dec 11 mock follows completion of every official subject clause. This distinction matters for the integrity of the claim "full syllabus".

ROUND 2 IMPLEMENTATION STATUS

Implemented: system of linear equations; socket API; other graphs representing data; numerical estimation.

Implemented: test-scope carry-through for these clauses in the relevant July-October assessment windows and later cumulative/full-syllabus tests.

Implemented: December Mock 1 and Mock 2 labels changed conceptually to pre-completion full-format mocks; Dec 11 marked as the first literal 100% syllabus mock.

Preserved: no existing subject progression, question counts, mock dates or revision cadence were removed.

ROUND 3 - DEPTH AUDIT NOW STARTS

Round 3 question: when the strategy says a topic is present, does it contain enough concrete study content to demonstrate substantive coverage of that official clause, rather than merely naming it?

Area	Depth finding	Classification
Engineering Mathematics	Most official clauses have concrete subtopics, but some are represented only by names in the dated schedule and gain detail only in mastery/memory layers.	Inspect in A3
Digital Logic	Minimization now names algebraic/K-map/tabular methods; depth of the actual tabular method mechanics and Boolean-algebra minimization procedure still needs verification.	Inspect in A3
COA	Control-unit and memory-interfacing corrections are explicit, but the schedule-level content is still compact compared with the design language of the official source.	Inspect in A3
Programming in C	Official source gives a broad "Programming in C" clause; strategy names several C fundamentals but the actual depth/skill inventory needs a dedicated audit.	Inspect in A3
Algorithms / TOC / OS / DBMS / CN	Strong explicit subtopic lists exist, but A3 will test whether important official clauses are only labels or are backed by sufficient instructional detail and mastery criteria.	Inspect in A3
General Aptitude	Several GA clauses are now explicit, but depth of grammar drills, DI formats, estimation, and spatial practice must be checked against the strategy content itself.	Inspect in A3

ROUND 3 - DEPTH AUDIT: VERIFICATION RESULT

Audit 3 was not implemented in v10.2. The file only announced that a depth audit would start and listed areas for inspection; it did not add the missing depth, competency statements, or test-depth controls. The following corrections therefore constitute the actual A3 implementation.

Depth standard: an official clause counts as substantively covered only when the strategy states a concrete study action, competency, or reconstruction task for that clause. A bare label such as "Linear Algebra", "Programming in C", or "performance" is not sufficient under this audit.

A3 GLOBAL DEPTH RULE - APPLIES TO EVERY SUBJECT

Study-depth rule: every official clause must have at least one explicit content item and one observable competency/task in the strategy.

Assessment-depth rule: every official clause must be represented in at least one scheduled assessment after it is taught; completed-subject revision tests and full-syllabus mocks must inherit the clause ledger.

Mastery-depth rule: where the plan already has a subject mastery section, the official clauses below are mapped to a concrete action rather than relying only on the high-level subject label.

No inflation rule: extra material may strengthen preparation, but it is not counted as evidence that an official clause was covered unless the official clause itself is explicitly represented.

A3 SUBJECT-BY-SUBJECT CORRECTIONS

Subject	A3 finding in v10.2	Binding depth correction
Engineering Mathematics	Several official clauses were named but only a subset had concrete mastery actions.	Add clause-level competency tasks for logic, sets/relations/functions/partial orders/lattices, monoids/groups, graph subclauses, combinatorics, matrices/determinants/systems/LU, calculus subclauses, and every named probability/distribution/statistics item.
Digital Logic	Methods were named, but method mechanics were not consistently made observable.	Add explicit practice of Boolean-algebra minimization, K-map minimization, tabular-method minimization, combinational/sequential circuit design, and fixed/floating-point arithmetic conversion.
COA	Official design/performance clauses were explicit after A1, but depth tasks remained compact.	Add separate competency tasks for instruction/addressing, ALU design, hardwired vs microprogrammed control, memory interfacing, hierarchy performance, cache mapping, interrupt/DMA, pipelining and hazards.
Programming & Data Structures	C had several fundamentals, but the broad "Programming in C" clause lacked a sufficiently explicit competency layer.	Add a C-programming competency block covering syntax/control-flow implementation, pointers/address reasoning, arrays, and implementation/debugging drills; retain all official data-structure subtopics as separate observable tasks.

ROUND 3 - DEPTH CORRECTIONS | ALGORITHMS TO OPERATING SYSTEMS

Subject	Binding depth implementation
Algorithms	Searching: perform representative search traces and compare outcomes. Sorting: trace and implement standard sorting procedures under timed conditions. Hashing: perform collision-resolution exercises. Complexity: derive worst-case time/space classification for listed algorithms. Greedy: solve a design problem and justify the choice. Dynamic programming: define state/transition and derive recurrence. Divide-and-conquer: write recurrence and identify the split/combine structure. Graph traversals: execute BFS/DFS. MST: construct and justify an MST. Shortest paths: execute and verify shortest-path calculations.
Theory of Computation	Regular expressions/finite automata: construct or trace the corresponding automaton. CFG/PDA: construct and trace derivations or stack behavior. Regular/CF languages: classify language types and connect them to the appropriate model. Pumping lemma: execute the proof workflow. Turing machines/undecidability: trace machine behavior and distinguish decidability status.
Compiler Design	Lexical analysis: identify tokenization/lexical structure. Parsing: construct/trace parse trees and parsing decisions. Syntax-directed translation: trace attribute/action flow. Runtime environments: trace the environment information used during execution. Intermediate code generation: produce/trace intermediate representations. Local optimization: apply an optimization example. Data-flow analysis: explicitly practise constant propagation, liveness, and common-subexpression elimination.
Operating Systems	System calls/processes/threads/IPC: trace interactions and identify the relevant execution entity. Concurrency/synchronization: reason about shared-state execution and synchronization. Deadlock: identify conditions and work through detection/prevention/avoidance examples. CPU/I/O scheduling: calculate scheduling outcomes. Memory management/virtual memory: trace allocation/address translation/page behavior. File systems: reason through file-system operations and structures.

These corrections do not replace the existing mastery criteria; they make the official clauses independently observable inside the same mastery/retrieval system.

A3 ASSESSMENT-DEPTH CONTROL

For every dated weekly/revision/monthly test, the scope now inherits the official clause ledger for every subject named in that test.

A test labelled by percentage (20%, 40%, 60%, 75%, 80%, 90%, or 100%) is not considered depth-complete unless the underlying official clauses assigned to that percentage have been explicitly taught and are eligible for questions.

For every completed subject, subsequent revision tests and full-syllabus mocks must retain representation of the complete clause set; a later broad subject label cannot silently drop a sub-clause.

December Mock 3 and every January mock are governed by the complete clause ledger because they occur after the claimed 100% completion point.

ROUND 3 - DEPTH CORRECTIONS | DATABASES, NETWORKS, GENERAL APTITUDE

Subject	Binding depth implementation
Databases	ER model: construct/interpret an ER representation. Relational model: map concepts into relations. Relational algebra: write equivalent expressions for query requirements. Tuple calculus: formulate tuple-based query expressions. SQL: solve query-writing tasks. Integrity constraints: identify and enforce required constraints. Normal forms: perform normalization/identify normal-form status. File organization: compare/trace organization behavior. Indexing: trace B/B+ tree index operations. Transactions/concurrency control: trace transaction schedules and control behavior.
Computer Networks	Layering: map functions to layers and trace encapsulation. Switching: distinguish circuit/packet/virtual-circuit behavior and compare performance implications. Performance metrics: calculate/interpret latency, bandwidth, throughput, delay, and utilization. Data link: solve error-detection/MAC/Ethernet tasks. Routing: trace distance-vector and link-state behavior. IPv4: work fragmentation/CIDR/NAT cases. TCP: solve flow-control and congestion-control cases. Socket API: trace client/server/socket interaction and addressing. DNS/HTTP: trace request/response flow and protocol purpose.
General Aptitude	Grammar: identify/correct each explicitly named grammar component. Vocabulary: solve words/idioms/phrases in context. RC/narrative sequencing: timed passage and ordering drills. DI: practise bar, pie, other graphs, 2D/3D plots, maps, tables. Numerical computation/estimation: timed ratio/percentage/power/exponent/log/P&C/series plus estimation exercises. Mensuration/geometry: direct calculation and figure-based drills. Elementary statistics/probability: compute/interpret basic statistical and probability questions. Analytical logic: deduction/induction, analogy, numerical relations/reasoning. Spatial: timed translation/rotation/scaling/mirroring/assembling/grouping plus folding/cutting/2D/3D pattern tasks.

A3 DEPTH GAPS THAT WERE CONFIRMED

Gap class	Status before A3	Status after A3
Broad EM labels	Present, but not all official clauses had observable actions	Corrected with clause-level competency map
DL minimization methods	Named, but mechanics not explicit	Corrected
COA design/performance	Named, but design clauses compact	Corrected
C programming depth	Fundamentals present, competency layer thin	Corrected
Algorithmic clause depth	Generally strong; formalized into clause actions	Hardened
TOC/Compiler/OS depth	Strong but previously dependent on subject-level labels in some places	Hardened
DBMS/CN depth	Strong subtopics, but not every official clause had an explicit observable task	Hardened
GA depth	Many clauses present, but drill specificity was uneven	Corrected

ROUND 3 - TEST / MOCK DEPTH VERIFICATION

Audit 3 also checks whether the assessment system actually tests the content deeply enough. This is not a demand that every test contain every syllabus clause; early tests are intentionally partial. The requirement is that a taught clause receives an appropriate assessment touchpoint and that cumulative testing does not erase it.

Assessment phase	Depth requirement	Implementation status
July staged tests	Each newly introduced clause must be represented in the weekly/revision scope, not just its parent subject.	Implemented via clause-ledger inheritance.
August staged tests	Stage-2 additions must have a named competency task or practice target; revision tests include earlier clauses.	Implemented.
September staged tests	TOC/COA completion and CN/OS additions must remain explicit at clause level.	Implemented.
October staged tests	DBMS/CN/OS completion blocks must carry the official clause set; percentage labels cannot substitute for clause mapping.	Implemented.
November completion tests	Digital Logic and final EM/DSA/GA completion must explicitly preserve every official clause.	Implemented.
December mocks	Mock 1/2 remain pre-completion full-format; Mock 3 is first literal 100% syllabus mock. Every completed clause is eligible thereafter.	Implemented/inherited from A2.
January daily mocks	Every mock uses the complete official clause ledger; emphasis zones cannot cause other official clauses to disappear.	Implemented as a binding test rule.

A3 result: The strategy now has an explicit depth layer connecting every official syllabus clause to an observable study competency and an assessment inheritance rule. The corrections are additive and do not remove the existing schedule.

ROUND 4 - SCHEDULE-PLACEMENT AUDIT NOW STARTS

Round 4 asks a different question: even when a clause exists and has adequate depth, is it placed in the correct chronological study window and then carried forward through revision/tests without timing contradictions?

Initial A4 check	Finding
July 12 Linear Algebra	System of linear equations is correctly attached after A2; verify all July-12 linear algebra subclauses are actually represented in the test scope.
July 25 GA grammar	A1 corrected clauses are present, but the exact GA 20% stage still needs chronological verification against every clause added under the grammar block.
Aug 9 COA	Hardwired/microprogrammed control was inserted into the August control-unit block; verify that the preceding and following COA tests do not imply completion without it.
Sep 6 COA	Memory interfacing/performance placement was inserted; verify it precedes the September COA-consolidation tests and is not omitted from later cumulative tests.
Sep-Oct GA/CN	Other graphs, estimation and socket API have explicit placements; A4 must verify their chronological handoff from teaching -> revision -> monthly test -> later mocks.
Nov 24 / Nov 26	The strategy marks several subjects 100% complete; A4 will reconstruct each completion claim chronologically rather than accepting the percentage label.
Dec 11 completion point	A4 will verify that every official clause has a legitimate pre-Dec-11 teaching placement before the first literal 100% mock.

Audit 4 is now active. No A4 schedule rewrites are being silently inserted in this version; the next pass will inspect every dated test window against the clause ledger before modifying any placement.